

MATHEMATICS X

Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF MATHEMATICS.

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QUESTIONS		Ex. No.	YEARS
UNIT 17:	SETS AND FUNCTIONS		
Section "B"			
2. If $A = \{1, 2, 3, 4\}$, then find the power set A .	17.1	2025	
2. If $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8, 10\}$ and $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ Prove that $(A \cap B)' = A' \cup B'$	17.3	2024	
2. If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8\}$ Prove that $(A \cup B)' = A' \cap B'$	17.3	2023	
Section "C"			
UNIT 18:	VARIATIONS		
Section "B"			
3. If $a:b = c:d = e:f$ then show that: $\frac{a^2b + c^2d + e^2f}{ab^2 + cd^2 + ef^2} = \frac{a + c + e}{b + d + f}$	18.6	2025	
3. Solve the equation by using componendo-dividendo theorem: $\frac{(x + 3)^2 + (x - 1)^2}{(x + 3)^2 - (x - 1)^2} = \frac{5}{4}$ [EXAMPLE:4]	18.4	2024	
3. If $a:b = c:d = e:f$ then show that: $\frac{a^4b^2 + a^2e^2 - e^4f}{b^6 + b^2f^2 - f^5} = \frac{a^4}{b^4}$	18.6	2023	

Section "C"		
UNIT 19:	MATRICES AND DETERMINANTS	
Section "B"		
Section "C"		
12. Find the inverse of the matrix by adjoint method: $A = \begin{bmatrix} 1 & 0 & 1 \\ -4 & 1 & -1 \\ 6 & -2 & 1 \end{bmatrix}$	19.2	2025
12. Find the invers of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$ by adjoint method	19.2	2024
12. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ -4 & 1 & -1 \\ 6 & -2 & 1 \end{bmatrix}$ by adjoint method.	19.2	2023

UNIT 20:	THEORY OF QUADRATIC EQUATIONS		
Section "B"			
9. Find the value of m , if the roots of the equation $x^2 - 5x + 2m = 0$ differ by 1.	20.3	2025	
5. Find all the cube roots of -125	20.2	2024	
5. If α, β are the roots of the equation $px^2 + qx + q = 0, p \neq 0$, find the value of $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}}$	20.4	2023	
Section "C"			

UNIT 21:	PARTIAL FRACTIONS		
Section "B"			
7. Resolve $\frac{11-3x}{(x-1)(x+3)}$ into partial fraction.	21.1	2025	
[EXAMPLE: 1]			
6. Resolve into partial fractions: $\frac{4x-3}{(x+1)^2}$	21.2	2024	

6. Resolve into partial fractions:	$\frac{4(x-4)}{x^2-2x-3}$	21.1	2023
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Section "C"

UNIT 22:	BASIC STATISTICS
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Section "B"

10. The marks obtained by students in a subject (out of 50) are given in the following grouped table. Find Mode.	22.3	2025
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Marks	25	30	35	40	45
	- 29	- 34	- 39	- 44	- 49
No. of Students	9	18	35	17	5

11. -- (OR)-- Find variance of number of absentees in a class for last seven days:	22.5	
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3, 5, 3, 2, 4, 1, 8

7. The following data show number of devices resulting in observation values in appropriate ranges. Find the Harmonic Mean (H.M.)	22.3	2024
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Class limit	10.5	11.0	11.5	12.0	12.5
	- 10.9	- 11.4	- 11.9	- 12.4	- 12.9
Frequencies	2	7	10	12	8

11. --OR-- Find the mean deviation of the marks of students which are 4, 7, 12, 76, 48, 4, 51.	22.5	
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7. The marks obtained by students in a subject (out of 50) are given in the following grouped table. Find Arithmetic Mean (A.M.).	22.3	2023
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Marks	25	30	35	40	45
	- 29	- 34	- 39	- 44	- 49
No. of Students	9	18	35	17	5

11. (OR) Find the standard deviation (S.D.) of the marks of students which are 90, 30, 85, 50, 25	22.5	
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[EXAMPLE: 1]

Section "C"

UNIT 23:		PYTHAGORAS' THEOREM	
Section "B"			
Section "C"			
14. In a right-angled triangle, the squares of the length of hypotenuse is equal to sum of the square of the length of the other two sides. Prove it. [THEOREM: 23.1]	23.1	2025	
14. If a square of one side of triangle is equal to the sum of the squares of the other two sides, then the triangles a right angled triangle. Prove that. [THEOREM: 23.2]	23.1	2024	
13. In a right angled triangle, the square of the length of hypotenuse is equal to sum of the squares of the length of the other two sides. Prove it. [THEOREM: 23.1]	23.1	2023	

UNIT 24:		RATIO AND PROPORTION	
Section "B"			
Section "C"			
13. If two triangles are similar, the measures of their corresponding sides are proportional. Prove it. [THEOREM: 24.4]	24.2	2025	
16. The internal bisectors of an angle of a triangle divides the sides opposite to it in the ratio of the sides containing the angle. Prove it. [THEOREM: 24.3]			
13. A line parallel to one side of a triangle and intersecting the other two sides, divides them proportionally. Prove it. [THEOREM: 24.1]	24.1	2024	
14.If a line segment intersects the two sides of a triangle in the same ratio then it is parallel to the third side. Prove it. [THEOREM: 24.2]	24.1	2023	

UNIT 25:		CHORDS OF A CIRCLE	
Section "B"			
5. If two chords of a circle are congruent, then they will be equidistant from the centre. Prove it.	25.4	2025	
9. A straight line, drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord. Prove it [THEOREM: 25.2]	25.1	2024	
9. A straight line, drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord. Prove it. [THEOREM: 25.2]	25.1	2023	
Section "C"			

UNIT 26:		TANGENTS OF A CIRCLE	
Section "B"			
Section "C"			
16 (OR). If two circles touch internally, the distance between their centres is equal to the difference of their radii. Prove it. [THEOREM: 26.2 CASE(B)]	26.1	2025	
16. If two circles touch externally, the distance between their centres is equal to the sum of their radii. Prove it. [THEOREM: 26.4 CASE(A)]	26.1	2024	
16 (OR). If a line is drawn perpendicular to a radial segment of a circle at its outer end point, it is tangent to the circle at that point. Prove it. [THEOREM: 26.1]			
16. If two circles touch internally, the distance between their centres is equal to the difference of their radii. Prove it. [THEOREM: 26.2 CASE(B)]	26.1	2023	
16.(OR) If a line is drawn perpendicular to a radial segment of a circle at its outer end point. it is tangent to the circle at that point. Prove it.			

[THEOREM: 26.1]			
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UNIT 27:	CHORDS AND ARCS
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Section "B"			
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8. If the angles subtended by two chords of a circle (or congruent circles) at the centres (corresponding centres) are equal, the chords are equal, Prove it. [THEOREM 27.4]	27.1	2025
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11. If two arcs of a circle are congruent then the corresponding chords are equal. Prove it. [THEOREM 27.1]	27.1	
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10. If two chords of a circle (or of congruent circles) are equal, then their corresponding arcs (minor, major, semi-circle) are congruent. Prove it, [THEOREM 27.2 (CONVERSE OF THEOREM 1)]	27.1	2024
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10. If two arcs of a circle (or congruent circles) are congruent then the corresponding chord are equal. Prove it. [THEOREM: 27.1]	27.1	2023
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Section "C"			
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UNIT 28:	ANGLE IN A SEGMENT OF A CIRCLE
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Section "B"			
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11. Any two angles in the same segments of a circle are equal. Prove it [THEOREM: 28.2]	28.1	2024
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11. The central angle of a minor arc is double in measure of the inscribed angle of the corresponding major arc. Prove it. [THEOREM: 28.1]	28.1	2023
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Section "C"			
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UNIT 29: PRACTICAL GEOMETRY-CIRCLES			
Section "B"			
Section "C"			
15. Draw an inscribed circle of $\triangle ABC$ in which $m\overline{AB} = 7\text{cm}$, $m\angle B = 60^\circ$ and $m\overline{BC} = 6\text{cm}$. (Also write steps of construction) [TOPIC 29.2(ii), Example Page 246]	29.2	2025	
15. Draw the circumcircle of $\triangle ABC$ in which $m\overline{AB} = 5.5\text{cm}$, $m\overline{AC} = 6\text{cm}$ and $m\angle A = 60^\circ$ (Also write steps of construction)	29.2	2024	
15. Draw two unequal circle of radii 3.4 cm and 2.1cm with centres A and B respectively such that $m\overline{AB} = 7.5\text{cm}$. Draw direct common tangents to these circles (Also write steps of construction).	29.5	2023	

UNIT 30: INTRODUCTION TO TRIGONOMETRY			
Section "B"			
4. Prove that $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \frac{\sin\theta}{1-\cos\theta}$ ($\cos\theta \neq 1$)	30.4	2025	
6. A ladder makes an angle of 60° with the ground and reaches a height of 6cm on the wall. Find the length of the ladder.	30.5		
4. Find the remaining trigonometric function if $\sin\theta = \frac{\sqrt{3}}{2}$ and θ lies in second quadrant.	30.3	2024	
8. Prove that $\sin^2\theta = \frac{\tan^2\theta}{1+\tan^2\theta}$	30.4		
4. Prove that $\frac{\cot\theta + \operatorname{cosec}\theta}{\sin\theta + \tan\theta} = \operatorname{cosec}\theta \cot\theta$	30.4	2023	
8. A ladder makes an angle 60° with the ground and reaches a height of 6 m on the wall, Find the length of the ladder.	30.5		
Section "C"			